Class: Name: Date:

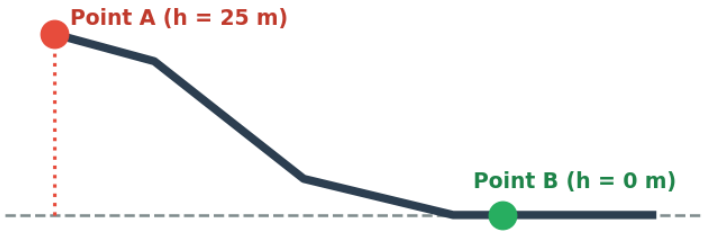
Kinetic & Potential Energy Calculations

EQUATION & VARIABLE REFERENCE SHEET

- Gravitational Potential Energy: GPE = m × g × h (m in kg, g = 9.8 m/s² unless stated, h in m, GPE in Joules)
- Kinetic Energy: KE = ½ × m × v² = 0.5 × m × v² (m in kg, v in m/s, KE in Joules. Note: velocity is squared)
- Conservation of Mechanical Energy: Total Energy = KE + GPE = constant (in frictionless / ideal systems)

Question 1: Conservation of Energy on a Track

A 600 kg roller coaster cart sits momentarily at rest at Point A (height = 25 m) before rolling down a frictionless track to Point B at ground level (height = 0 m). Assuming g = 9.8 m/s² and zero losses to friction, what is the kinetic energy of the cart when it reaches Point B?



[ ] A. 110,250 J
[ ] B. 73,500 J
[ ] C. 147,000 J
[ ] D. 0 J
[ ] E. 15,000 J
[ ] F. 294,000 J

Workspace / Calculation:

Question 2: Free Fall Impact Velocity

A 6 kg bowling ball is dropped from rest from the top of a 5-meter-high vertical cliff. Neglecting air resistance and using an approximated g = 10 m/s², what is the speed of the bowling ball just before it strikes the ground?

[ ] A. 10 m/s
[ ] B. 5 m/s
[ ] C. 20 m/s
[ ] D. 15 m/s
[ ] E. 100 m/s
[ ] F. 5 m/s

Class: Name: Date:



Workspace / Calculation:

Question 3: Kinetic Energy Calculation

A remote-controlled toy racing car with a mass of 2.0 kg drives down a straight, flat track at a constant velocity of 8 m/s. What is its kinetic energy (KE)?

[ ] A. 96 J
[ ] B. 8 J
[ ] C. 128 J
[ ] D. 128 J
[ ] E. 16 J
[ ] F. 64 J

Workspace / Calculation:

Question 4: Rearranging KE to Solve for Velocity

A ball of mass 0.5 kg possesses exactly 225 J of kinetic energy as it travels horizontally through the air. What is the instantaneous velocity of the ball?

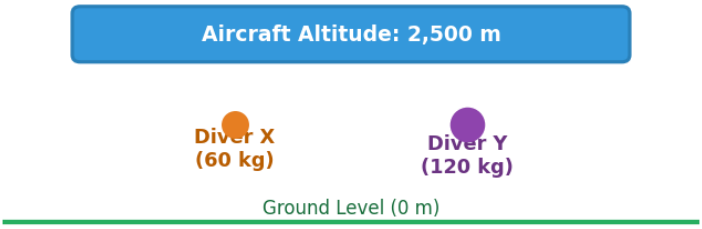
[ ] A. 60 m/s
[ ] B. 15 m/s
[ ] C. 15.0 m/s
[ ] D. 30 m/s
[ ] E. 45 m/s
[ ] F. 450 m/s

Workspace / Calculation:

Question 5: Mass Proportionality of Potential Energy

Two skydivers, Diver X (60 kg) and Diver Y (120 kg), stand inside an aircraft preparing to jump from an elevation of 2,500 meters above the landing zone. How does the gravitational potential energy of Diver Y compare to Diver X prior to jumping?

Class: Name: Date:



[ ] A. Diver Y has 2 times the GPE of Diver X because GPE is directly proportional to mass.

[ ] B. Diver Y has less GPE because heavier objects experience less gravity.

[ ] C. Diver Y has half the GPE of Diver X.

[ ] D. Both divers have the exact same GPE because they are at the identical altitude.

[ ] E. Diver Y has 4 times the GPE of Diver X.

[ ] F. Neither diver has any GPE until they begin moving downward.

Workspace / Calculation:

Question 6: Velocity Scaling Effect on Kinetic Energy

If a motor vehicle increases its speed by a factor of 4 (from 10 m/s to 40 m/s), by what multiplier does its kinetic energy (KE) increase?

[ ] A. It doubles (2 times greater)

[ ] B. It remains completely unchanged

[ ] C. It increases by 16 times (4^2)

[ ] D. It increases by 8 times

[ ] E. It increases by 32 times

[ ] F. It increases by 4 times

Workspace / Calculation:

Question 7: Gravitational Potential Energy Calculation

A construction worker places a 2 kg brick on top of a 6-meter-high scaffolding. Using $g = 9.8\ m/s^2$, what is the gravitational potential energy (GPE) of the brick relative to the ground?

[ ] A. 235 J

[ ] B. 120 J

[ ] C. 59 J

[ ] D. 176 J

[ ] E. 12 J

[ ] F. 117.6 J

Class: Name: Date:



Workspace / Calculation:

Question 8: Rearranging GPE to Solve for Height

An overhead construction crane lifts a 200 kg steel girder straight up, conferring 23,520 J of gravitational potential energy relative to the ground. Taking g = 9.8 m/s², what vertical height was the beam lifted?

[ ] A. 24 meters
[ ] B. 23.5 meters
[ ] C. 12 meters
[ ] D. 6.0 meters
[ ] E. 18.0 meters
[ ] F. 9.0 meters

Workspace / Calculation:

Question 9: Spring Potential to Kinetic Energy Transfer

A student compresses a spring launcher horizontally to propel a 0.5 kg steel ball along a smooth, frictionless lab table. If the released spring transfers 16.0 J of elastic potential energy entirely into kinetic energy of the ball, what speed does the ball achieve?

[ ] A. 32.0 m/s
[ ] B. 12.0 m/s
[ ] C. 4.0 m/s
[ ] D. 16.0 m/s
[ ] E. 8.0 m/s
[ ] F. 16.0 m/s

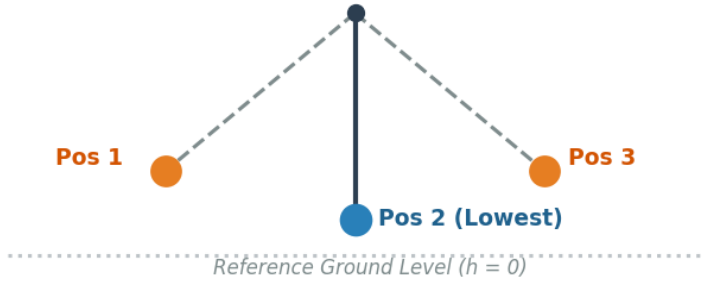
Workspace / Calculation:

Class: Name: Date:



Question 10: Energy Transformations in a Pendulum

Examine the simple pendulum swinging continuously between positions 1, 2, and 3 without air resistance. Which statement accurately describes the energy state of the pendulum bob at Position 2 (the lowest point of the swing)?



[ ] A. GPE has increased compared to Position 1.

[ ] B. Total mechanical energy has doubled compared to Position 1.

[ ] C. The pendulum has completely stopped moving.

[ ] D. GPE is at its maximum and KE is at its minimum.

[ ] E. KE is at its maximum and GPE is at its minimum.

[ ] F. Both KE and GPE are equal to zero.

Workspace / Calculation: